

Bayes' Theorem

Definition

Bayes' Theorem is a mathematical rule used to **update the probability of an event when new evidence becomes available.**

In simple terms:

Bayes' Theorem tells us how to revise our belief based on new information.

For example, if a medical test result is positive, Bayes' Theorem helps determine the probability that the person actually has the disease, considering both the test accuracy and the disease's prevalence.

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Where:

- $P(A)$ = Prior Probability (initial belief)
- $P(B)$ = Probability of the evidence
- $P(B|A)$ = Likelihood (probability of observing the evidence if A is true)
- $P(A|B)$ = Posterior Probability (updated belief after seeing the evidence)

Components of Bayes' Theorem

Term	Meaning	Example
Prior Probability	Initial belief before evidence	1% of people have a disease
Likelihood	Chance of observing evidence if the event is true	Test detects disease 99% of the time
Evidence	New information obtained	Test result is positive
Posterior Probability	Updated probability after considering evidence	Probability the patient actually has the disease

Spam Email Filter

Scenario

Imagine Gmail receives **100 emails** in one day.

Out of these:

- **20 emails are Spam**
- **80 emails are Genuine (Not Spam)**

Now, suppose the word "**Congratulations**" appears:

- In **14 spam emails**
- In **8 genuine emails**

Step 1: Understand the Situation

Email Type	Total Emails	Contains "Congratulations"
Spam	20	14
Genuine	80	8
Total	100	22

Prior probability:

$$P(\text{Spam}) = \frac{20}{100} = 0.20$$

Likelihood:

$$P(\text{Congratulations} | \text{Spam}) = \frac{14}{20} = 0.70$$

Total probability of seeing "Congratulations":

$$P(\text{Congratulations}) = \frac{22}{100} = 0.22$$

Therefore,

$$P(\text{Spam} | \text{Congratulations}) = \frac{0.70 \times 0.20}{0.22} = 0.636$$

63.6%

Interpretation

Initially, we believed there was only a **20% chance** that the email was spam.

After seeing the word "**Congratulations**", our belief changes.

Now the probability that the email is spam becomes **63.6%**.

This updated probability is called the **Posterior Probability**.

Bayes' Theorem Flow

None

Before seeing the email



20% chance it is Spam



Evidence:

Email contains "Congratulations"



Apply Bayes' Theorem



Updated probability

63.6% chance it is Spam

Bayes' Theorem Example: Medical Test (Using Counts)

Scenario

A hospital tests **1,000 people** for a disease.

Out of these:

- **10 people actually have the disease.**
- **990 people are healthy.**

Before conducting the test, what is the probability that a randomly selected person has the disease?

Answer:

$$P(\text{Disease}) = \frac{10}{1000}$$

This is called the **Prior Probability**.

Step 1: Test Accuracy

The medical test works as follows:

- Out of **10 diseased people**, the test correctly identifies **9** as positive.
- Out of **990 healthy people**, the test incorrectly identifies **50** as positive (false positives).

This gives the following table:

Person's Condition	Total People	Test Positive	Test Negative
Diseased	10	9	1
Healthy	990	50	940
Total	1000	59	941

Step 2: New Evidence

Suppose a person's **test result is Positive**.

Now we ask:

What is the probability that this person actually has the disease?

Prior Probability

Out of 1000 people,

$$P(\text{Disease}) = \frac{10}{1000}$$

Likelihood

Among the 10 diseased people,
9 tested positive.

$$P(\text{Positive} | \text{Disease}) = \frac{9}{10}$$

Probability of a Positive Test

A total of **59 people** tested positive.

$$P(\text{Positive}) = \frac{59}{1000}$$

Bayes Calculation

$$P(\text{Disease} | \text{Positive}) = \frac{\frac{9}{10} \times \frac{10}{1000}}{\frac{59}{1000}} = \frac{9}{59} = 0.1525$$

Even Simpler Way (No Formula)

Ask your students to look only at the people who tested positive.

From the table:

- **9 people actually have the disease.**
- **50 people are healthy but still tested positive.**

So, the total number of positive test results is:

$$9 + 50 = 59 \quad 599 + 50 = 599 + 50 = 59$$

Among these **59 positive results**, only **9 people truly have the disease**.

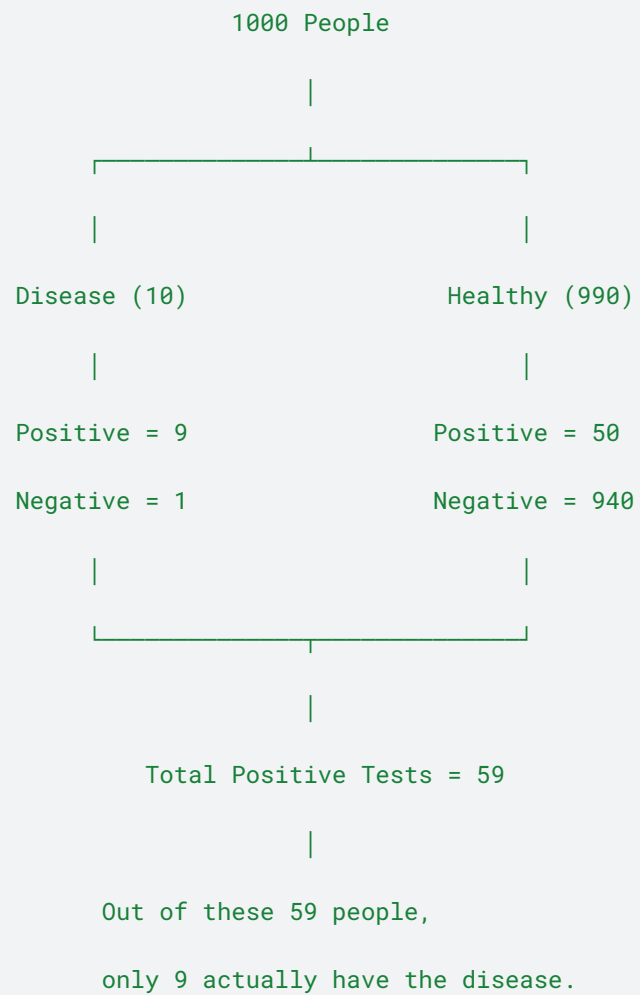
Therefore,

$$P(\text{Disease} | \text{Positive}) = \frac{9}{59} \quad P(\text{Disease} | \text{Positive}) = \frac{9}{599}$$

This gives the same answer as Bayes' Theorem.

Visual Explanation

None



Probability Distributions

What is a Probability Distribution?

A **probability distribution** describes how the possible values of a random variable are distributed and the probability associated with each value.

In simple terms,

It tells us how likely different outcomes are.

Example

Example

When rolling a fair die:

Outcome	Probability
1	$1/6$
2	$1/6$

3	1/6
4	1/6
5	1/6
6	1/6

Types of Probability Distributions

Probability distributions are broadly classified into two categories.

1. Discrete Probability Distribution

A discrete probability distribution deals with countable values.

Examples:

- Number of heads in 5 coin tosses

- Number of defective bulbs
- Number of students absent

Common distributions:

- Binomial Distribution
- Poisson Distribution

2. Continuous Probability Distribution

A continuous probability distribution deals with measurable values.

Examples:

- Height
- Weight
- Temperature
- Salary

Common distribution:

- Normal Distribution

Binomial Distribution

Definition

A **Binomial Distribution** is used to find the probability of getting a specific number of **successes** in a **fixed number of independent trials**, where each trial has only **two possible outcomes**.

The two outcomes are usually:

- Success / Failure
- Pass / Fail
- Head / Tail
- Yes / No
- Defective / Non-defective

Conditions for Binomial Distribution

Before solving any problem, check these four conditions:

- ✓ Fixed number of trials (n)
- ✓ Two possible outcomes
- ✓ Probability remains constant
- ✓ Trials are independent

Binomial Probability Formula

$$P(x; n, p) = \binom{n}{x} p^x (1 - p)^{n-x}$$

Where:

x = number of occurrences of a specific outcome in n trials

p = probability of success in a single trial

n = number of trials

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$



www.inchcalculator.com

Example: Coin Toss

Problem

A fair coin is tossed **5 times**.

Find the probability of getting **exactly 3 Heads**.

Step 1: Check the Conditions

- Fixed trials → **5 tosses** ✓
- Two outcomes → **Head/Tail** ✓
- Probability remains constant → **0.5** ✓
- Independent tosses ✓

Therefore, this is a **Binomial Distribution**.

Step 2: Identify the Parameters

Number of trials - **$n=5$**

Number of successes - **$x=3$**

Probability of success - **$p=0.5$**

Probability of failure - **q=1-p=0.5**

Step 3: Formula

$$P(X = x) = \binom{n}{x} p^x q^{n-x}$$

Substitute the values:

$$P(X = 3) = \binom{5}{3} (0.5)^3 (0.5)^2$$

Step 4: Calculate Combination

$$\begin{aligned} \binom{5}{3} &= \frac{5!}{3!(5-3)!} \\ &= \frac{5!}{3!2!} \\ &= \frac{5 \times 4}{2 \times 1} \\ &= 10 \end{aligned}$$



Step 5: Calculate Probability

$$\begin{aligned}P(X = 3) &= 10(0.5)^3(0.5)^2 \\&= 10(0.5)^5 \\&= 10 \times 0.03125 \\&= 0.3125\end{aligned}$$

Final Answer

The probability of getting exactly **3 Heads** is

0.3125



Poisson Distribution

Definition

A **Poisson Distribution** is a discrete probability distribution that is used to **find the probability of a certain number of events occurring in a fixed interval of time, space, distance, or area**, when these events occur **randomly and independently** at a **constant average rate**.

In Simple Words

Poisson Distribution helps us answer the question: "How many times will an event happen in a given period or area?"

Unlike the **Binomial Distribution**, we are **not performing a fixed number of trials**. Instead, we are **counting how many events occur**.

Conditions for Poisson Distribution

A problem follows a Poisson Distribution if:

- ✓ We are counting the **number of events**.
- ✓ The events occur in a **fixed interval** (time, area, distance, etc.).
- ✓ Events occur **independently**.

- ✓ The **average rate (λ)** remains constant.

Parameter

Poisson Distribution has only **one parameter**.

λ (Lambda)

It represents the **average number of events** in a fixed interval.

For example:

- Average 5 customers per hour
- Average 20 emails per day
- Average 3 accidents per month

Formula

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Where:

- λ = Average number of events
- x = Number of events for which probability is required
- e = Euler's constant (≈ 2.718)
- $x!$ = Factorial of x

Example : Customer Arrivals

Problem

A supermarket receives **4 customers per minute on average**.

Find the probability that **exactly 2 customers** arrive in the next minute.

Step 1: Check the Conditions

- Counting customers ✓
- Fixed time = 1 minute ✓
- Average rate = 4 customers/minute ✓
- Arrivals are independent ✓

Therefore,

Poisson Distribution

Step 2: Identify the Parameters

Average customers:

$$\lambda=4$$

Required customers:

$$x=2$$

Step 3: Apply the Formula

$$P(X = 2) = \frac{e^{-4} \times 4^2}{2!}$$

Step 4: Calculate

We know:

$$e^{-4} = 0.0183$$

$$4^2 = 16$$

$$2! = 2$$

Substitute:

$$\begin{aligned} P(X = 2) &= \frac{0.0183 \times 16}{2} \\ &= \frac{0.2928}{2} \\ &= 0.1464 \end{aligned}$$

Normal Distribution (Gaussian Distribution)

Introduction

A **Normal Distribution** is one of the most important probability distributions in statistics. It is a **continuous probability distribution** where the data is symmetrically distributed around the mean, forming a **bell-shaped curve**.

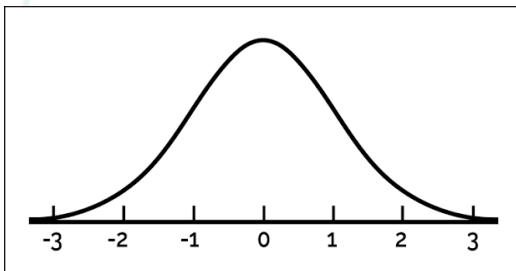
It is also known as the **Gaussian Distribution**, named after the mathematician Carl Friedrich Gauss.

Simple Definition

A Normal Distribution is a probability distribution in which most values are concentrated around the average (mean), and fewer values occur as we move away from the mean.

Why is it Called a Bell Curve?

When we plot normally distributed data on a graph, it forms a curve that looks like a bell.



- Highest point = Mean
- Both sides are identical
- Left side mirrors the right side

Real-Life Examples

Many real-world measurements approximately follow a normal distribution.

Examples include:

- Heights of people
- Weights of adults
- IQ scores
- Blood pressure
- Student exam marks
- Shoe sizes
- Manufacturing measurements
- Daily temperatures (approximately)

Characteristics of Normal Distribution

1. Bell-Shaped Curve

The graph always forms a bell shape.

- Highest frequency at the center
- Lowest frequency at both ends

2. Symmetrical Distribution

The left and right halves are mirror images.

3. Mean = Median = Mode

For a Normal Distribution:

None

Mean = Median = Mode

All three measures of central tendency are equal and lie at the center.

4. Continuous Distribution

Normal Distribution is used for **continuous data**.

Examples:

- Height = 165.3 cm
- Weight = 62.75 kg
- Temperature = 35.8°C

These values can take **any value within a range**.

5. Total Area Under the Curve = 1

The entire area under the curve represents **100% of the observations**.

This means every possible observation is included under the curve.

6. Tails Never Touch the X-Axis

The curve gets closer and closer to the x-axis but never actually touches it.

Parameters of Normal Distribution

A Normal Distribution is defined by **two parameters**.

1. Mean (μ)

The **mean** represents the center of the distribution.

It tells us where most observations are located.

Example:

Average marks = 70

Then,

$$\mu=70$$

2. Standard Deviation (σ)

The **standard deviation** measures how spread out the data is around the mean.

- Small $\sigma \rightarrow$ Data is closely clustered around the mean.
- Large $\sigma \rightarrow$ Data is more spread out.

Example

Class A:

Marks:

68, 69, 70, 71, 72

(Standard deviation is small.)

Class B:

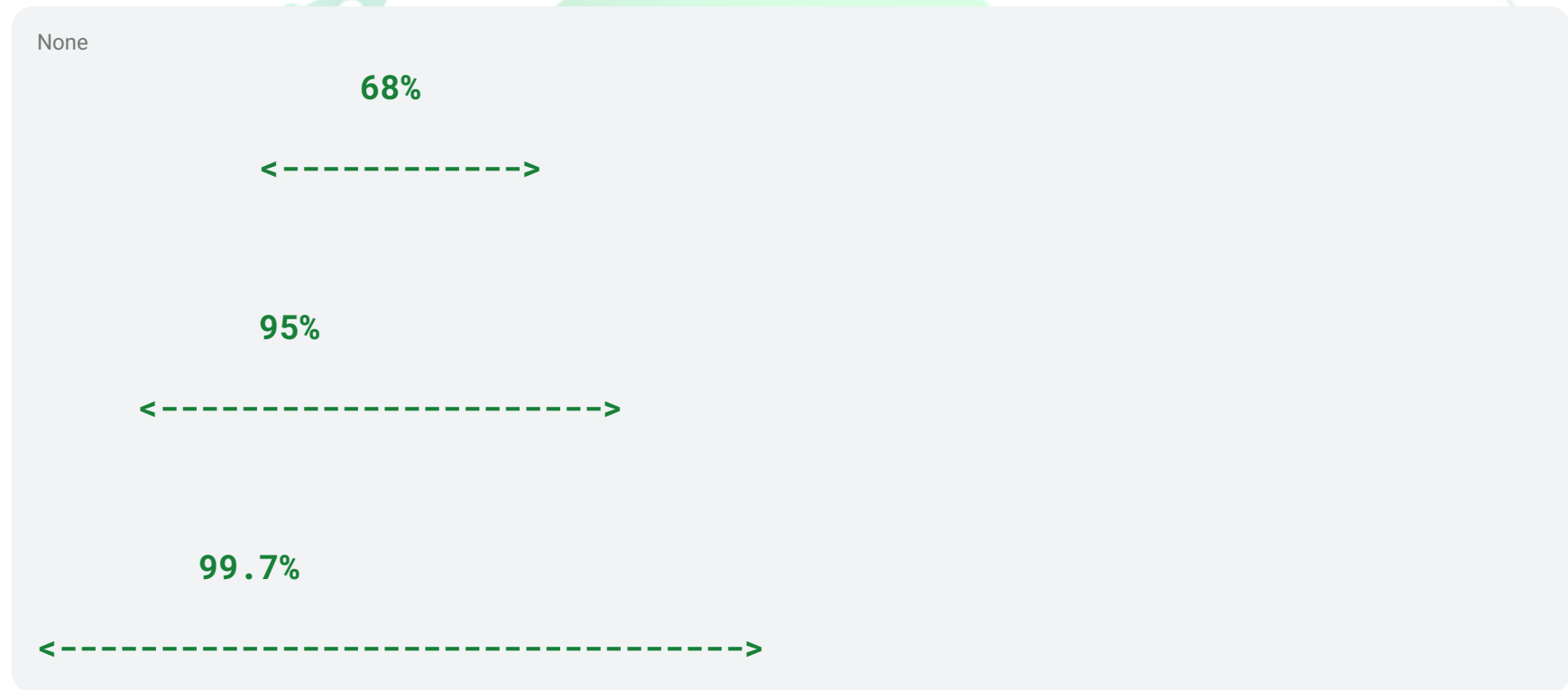
20, 40, 70, 90, 100

(Standard deviation is large.)

Although both classes have the same average (mean), the spread is very different.

Empirical Rule (68–95–99.7 Rule)

One of the most important properties of a Normal Distribution is the Empirical Rule.



Rule

- About 68% of observations lie within 1 standard deviation ($\mu \pm 1\sigma$).

- About 95% lie within 2 standard deviations ($\mu \pm 2\sigma$).
- About 99.7% lie within 3 standard deviations ($\mu \pm 3\sigma$).

Example

Suppose the marks of students have:

Mean = 70

Standard deviation = 10

Within 1 Standard Deviation

70 ± 10

= 60 to 80

About 68% of students score between 60 and 80.

Within 2 Standard Deviations

70 ± 20

= 50 to 90

About 95% of students score between 50 and 90.

Within 3 Standard Deviations

70 ± 30

= 40 to 100

About 99.7% of students score between 40 and 100.

Advantages

- Simple to understand and interpret.
- Describes many natural and real-world phenomena.
- Forms the foundation for many statistical methods.
- Widely used in machine learning and data analysis.

Limitations

- Not all datasets are normally distributed.
 - Sensitive to extreme outliers.
 - May not model highly skewed or multimodal data well.
-

How to Identify a Normal Distribution

Ask these questions:

1. Is the data **continuous**?
2. Is the graph **bell-shaped**?
3. Is it **symmetric** around the mean?
4. Are the **mean, median, and mode approximately equal**?
5. Are most values concentrated near the center?

If the answer is **Yes** to these questions, the data likely follows a **Normal Distribution**.

Memory Trick

- **Binomial:** Fixed number of trials, two outcomes.
- **Poisson:** Count the number of events in a fixed interval.
- **Normal:** Continuous data that clusters around an average, forming a bell-shaped curve.

Feature	Binomial Distribution	Poisson Distribution	Normal Distribution
Type	Discrete	Discrete	Continuous
Measures	Successes in fixed trials	Number of events in an interval	Continuous measurements
Number of Trials	Fixed	Not fixed	Not applicable
Outcomes	Two outcomes	Event counts	Infinite possible values
Shape	Varies with parameters	Right-skewed for small λ , more symmetric as λ increases	Bell-shaped and symmetric

Examples	Coin tosses, pass/fail	Calls per hour, customer arrivals	Heights, weights, exam scores
----------	---------------------------	-----------------------------------	----------------------------------

